

employment identifies:

$$\begin{aligned}\Delta_{\text{E-NE}}^N &= \mathbb{E}[\Delta \log N_j \mid E_j = 1] - \mathbb{E}[\Delta \log N_j \mid E_j = 0] \\ &= \frac{\epsilon - \gamma}{\epsilon - 1} \kappa - (1 - \gamma)(\alpha_{b0} + \alpha_{b1}) - \gamma(\alpha_{c0} + \alpha_{c1}),\end{aligned}\tag{45}$$

2. Workplace wage bill effect of employer encouragement. Following Equation (38), the difference-in-differences estimate of employer encouragement on the workplace wage bill identifies:

$$\begin{aligned}\Delta_{\text{E-NE}}^{WH} &= \mathbb{E}[\Delta \log W_j H_j \mid E_j = 1] - \mathbb{E}[\Delta \log W_j H_j \mid E_j = 0] \\ &= \kappa\end{aligned}\tag{46}$$

A.3 Technical Derivations

A.3.1 Missing Intercept

The difference-in-differences estimators in Equations (41)–(43) identify earnings effects up to the *skill price effect* in non-encouraged firms, $d \log W_0$. This is the familiar *missing intercept* problem: within-sector difference-in-differences do not capture general equilibrium (sector-wide) effects.

In this section, we use the model’s structure to characterize the residual term. The basic idea is that adopters generate spillovers to non-adopters only when their own market outcomes change. Thus, observing null effects of adoption on market outcomes at both the worker and workplace levels implies that spillovers to non-adopters are likewise limited.

To formalize this idea, we need to take a stance on how firms compete and product markets interact. We assume a nested CES demand function with elasticity η across and elasticity ϵ within. The product demand curve (Equation (14)) then becomes:

$$Y_j = P_j^{-\epsilon} P^{\epsilon-\eta},\tag{47}$$

where $P = (\int P_j^{1-\epsilon} dj)^{\frac{1}{1-\epsilon}}$ is the sectoral price index.

Combining this with Equations (10), (13), (15), and (47) implies that the skill price effect

in non-encouraged firms reflects two terms:

$$d \log W_0 = \underbrace{-\frac{1}{\epsilon} d \log Y_0}_{\text{Residual demand effect}} + \underbrace{\frac{\eta - \epsilon}{\eta} d \log Y}_{\text{Sector competition effect}}, \quad (48)$$

where the *residual demand effect* reflects that expanding firms move down their residual demand curves, and the *sector competition effect* reflects that competing firms expand by stealing sectoral demand.

The change in output of non-encouraged firms, $d \log Y_0$, reflects both productivity and employment responses:

$$d \log Y_0 = \underbrace{\mathbb{E}[A_{i0} B_{i0}]}_{\text{Adopter productivity}} + \underbrace{\mathbb{E}[d \log N_0]}_{\text{Employment effect}}, \quad (49)$$

where the *adopter productivity* and *productivity* effects are identified by:

$$\mathbb{E}[A_{i0} B_{i0}] = A_0 \times \Delta_{\text{NE}, \text{A-NE}, \text{NA}}^Y \quad (50)$$

$$\mathbb{E}[d \log N_0] = -E \times \Delta_{\text{E-NE}}^N, \quad (51)$$

where A_0 is the adoption rate in non-encouraged firms and E is the share of firms that encourage AI chatbots. Equation (51) utilizes the labor market clearing condition $E \times d \log N_1 + (1 - E) d \log N_0 = 0$.

Similarly, total output growth, $d \log Y$, reflects aggregate adopter productivity:

$$d \log Y = (1 - E) \times A_0 \times \Delta_{\text{NE}, \text{A-NE}, \text{NA}}^Y + E \times A_1 \times \Delta_{\text{E}, \text{A-E}, \text{NA}}^Y, \quad (52)$$

noting that total employment effects cancel in aggregate (due to labor market clearing).

Putting these together, the skill-price effect in non-encouraged firms is:

$$\mathbb{E}[d \log W_0] = -\frac{1}{\eta} \left[A_0 \cdot \Delta_{\text{NE}, \text{A-NE}, \text{NA}}^Y - E \times \Delta_{\text{E-NE}}^N \right] \quad (53)$$

$$+ \frac{\eta - \epsilon}{\eta} \left[(1 - E) A_0 \cdot \Delta_{\text{NE}, \text{A-NE}, \text{NA}}^Y + E A_1 \cdot \Delta_{\text{E}, \text{A-E}, \text{NA}}^Y \right], \quad (54)$$

where all terms are identified by our existing difference-in-differences estimates, up to

calibrated values of the demand elasticities η and ϵ .

Importantly, when both the firm-level employment difference-in-differences Δ^N and the worker-level earnings difference-in-differences Δ^Y are approximately zero (as in Figures 2 and 3), this implies that the missing intercept, $\mathbb{E}[d \log W_0]$, is also close to zero.

A.3.2 Heterogeneous Encouragement Costs

For analytical simplicity, our main analysis focused on the case where firms face the same encouragement cost, $\kappa_j = \kappa$ for all j . In this section, we consider the general case with heterogeneous encouragement costs, letting F_κ denote the CDF of employer encouragement costs.

When encouragement costs are heterogeneous, equilibrium skill price differentials reflect both the supply of and demand for human capital across encouraged and non-encouraged employers. These, in turn, depend on workers' sorting and adoption decisions, as well as firms' encouragement and hiring policies. We begin by characterizing these decision rules in terms of the relative skill price, $W_{1:0} = W_1/W_0$, and then combine supply and demand to solve for the resulting equilibrium skill differentials.

Employer encouragement. Following the cutoff rule in Equation (19), the share of employers encouraging AI chatbots satisfies:

$$\mathbb{P}[E_j = 1] = F_\kappa[(1 - \epsilon) \log W_{1:0}], \quad (55)$$

where F_κ denotes the CDF of encouragement costs.

Workplace labor demand. Encouraged firms gain a marginal productivity advantage through lower skill prices, increasing their demand for human capital by:

$$\log H_{1:0} = -\epsilon \log W_{1:0}. \quad (56)$$

Demand for human capital. Combining Equations (55)-(56), the aggregate demand for human capital by encouraged firms relative to non-encouraged firms satisfies

$$H_{1:0}^D(\underbrace{W_{1:0}}_{-} | \underbrace{F_{\kappa}}_{+}, \underbrace{\epsilon}_{+}) = \frac{\mathbb{P}[E_j = 1] H_{1:0}}{1 - \mathbb{P}[E_j = 1]} \quad (57)$$

$$= \frac{F_{\kappa} [(1 - \epsilon) \log W_{1:0}] W_{1:0}^{-\epsilon}}{1 - F_{\kappa} [(1 - \epsilon) \log W_{1:0}]}, \quad (58)$$

which is strictly decreasing in the relative skill price $W_{1:0}$, strictly increasing in the elasticity of product demand ϵ (since $W_{1:0} < 1$ in equilibrium), and increasing in F_{κ} . Intuitively, the relative demand for human capital at encouraged firms is higher when encouragement costs are low (i.e., $F_{\kappa}(x)$ is high for any x), the relative skill price $W_{1:0}$ is low (enhancing the productivity advantage of encouraged firms), and product demand is highly elastic (allowing firms to scale more easily, recoup encouragement costs, and hire more workers).

Adoption. The share of workers who sort into encouraged firms and adopt chatbots satisfies (following Equation (22)):

$$\mathbb{P}[A_i = 1] = i^* = -\frac{1}{\hat{\beta}}(\hat{\alpha}_0 + \hat{\alpha}_1 + \log W_{1:0}). \quad (59)$$

Supply of human capital. The aggregate supply of human capital to encouraged firms satisfies:

$$H_1^S(\underbrace{W_{1:0}}_{+} | \underbrace{\alpha_b}_{+}, \underbrace{\alpha_c}_{-}, \underbrace{\beta_b}_{+}, \underbrace{\beta_c}_{-}) = \int_0^{i^*} \exp(\alpha_{b0} + \alpha_{b1} + \beta_b i) di \quad (60)$$

$$= \frac{\exp(\alpha_{b0} + \alpha_{b1})}{\beta_b} [\exp(\beta_b i)]_0^{i^*} \quad (61)$$

$$= \frac{\exp(\alpha_{b0} + \alpha_{b1})}{\beta_b} \left[\exp(-\frac{\beta_b}{\hat{\beta}}(\hat{\alpha}_0 + \hat{\alpha}_1 + \log W_{1:0})) - 1 \right], \quad (62)$$

which is strictly increasing in the relative skill price ($W_{1:0}$), strictly increasing in both the common and idiosyncratic productivity benefits of AI chatbots (α_b and β_b), and strictly decreasing in both the common and idiosyncratic adoption costs (α_c and β_c)—under the assumption that chatbots are never counterproductive when encouraged (i.e., $B_{ij}(i =$